Yu-Jui Liang 梁栴睿

Algorithms / System Simulation / Testing & Validation

學歷：國立臺灣大學｜資料科學碩士學位學程（預計 2027 年 6 月畢業）
國立政治大學｜資訊科學系學士（雙主修：電子物理學士學位學程）
指導教授：江介宏教授（Jie-Hong Roland Jiang），國立臺灣大學
共同指導：鄭湘筠博士（Hsiang-Yun Cheng），中央研究院
專業領域：C++/Python, optimization, system simulation, GPU/memory systems, validation

01 / GPU / MEMORY SYSTEMS (ONGOING)

Latency- and Energy-Aware Scheduling for Heterogeneous GPU LLM Serving

Purpose. Study efficient LLM inference serving across heterogeneous GPUs with different compute capabilities, HBM / HBF memory configurations, and prefill / decode execution characteristics. The research focuses on resource imbalance, memory-access overhead, and inter-device data movement as potential sources of inference latency and energy inefficiency.

Approach. Designing a system-level simulation methodology to investigate request-to-GPU scheduling, prefill / decode placement, memory allocation, and data-movement policies under varying workloads. The planned evaluation compares scheduling and memory-system configurations using time to first token (TTFT), time per output token (TPOT), end-to-end latency, throughput, power consumption, and energy per generated token. The objective is to identify configurations and policies that improve latency and energy efficiency while respecting compute, memory-capacity, and bandwidth constraints.

Evidence / Status. Ongoing research focused on developing a system-level simulation framework for heterogeneous GPU LLM serving, with emphasis on latency-aware scheduling, memory-system optimization, and energy-efficient resource allocation.

Relevant skills: C/C++, Python, Linux, Algorithm Design, Optimization, System-Level Simulation, Performance Modeling, Resource Scheduling, Memory Management, Performance Analysis

GPU / Memory Configuration ⟶ Request Scheduling / Memory Policies ⟶ Latency / Energy Evaluation

02 / LLM APPLICATIONS & FORMAL REASONING

SAT-LEX: LLM and SMT-Based Document Verification System

Purpose. Support more deterministic checking of domain-specific documents by combining extracted evidence with explicit rules and automated constraint solving.

Approach. Translated extracted document evidence and domain rules into structured constraints; applied an SMT solver to identify conflicting conditions and unsupported conclusions beyond unconstrained LLM responses.

Evidence / Status. The implemented constraint checks are intended to identify logical conflicts, inconsistent conditions, and unsupported conclusions.

Relevant skills: LLM applications, SMT / constraint solving, automated reasoning, verification workflows

Rules / Document Evidence ⟶ Structured Constraints ⟶ Solver Checks

03 / BACKEND SYSTEMS & SOFTWARE VALIDATION

EV Charging Management System – NeoPower Technologies

Purpose. Develop and validate an electric-bus charging management platform integrating vehicles, charging equipment, and external vendor services, with reliable operation across customer sites.

Approach. Contributed to system planning and backend development, including APIs, database integration, and log-based debugging. Used a container-based architecture to isolate projects and services on shared servers, reducing interference during deployment and troubleshooting. Independently developed a custom simulator to emulate external vendors acting as clients or servers. Reconstructed normal and failure scenarios from historical database logs and customer feedback, including sudden charging-demand surges and simultaneous charger disconnections caused by site-network failures.

Evidence / Status. Professional engineering experience (Jun. 2023 – Mar. 2025). Used scenario-based simulation and regression testing before deployment, followed by validation using production logs, operational data, and customer feedback. Supported A/B-style comparisons of existing and revised algorithms across sites with similar conditions, such as charging-power allocation policies, to assess behavior under real operating conditions.

Relevant skills: Backend development, API / database integration, containers, vendor simulation, regression testing, production validation, log analysis, debugging

Vendor Simulation ⟶ Scenario / Regression Tests ⟶ Deployment ⟶ Operational Validation

04 / LOGIC SYNTHESIS & TECHNOLOGY MAPPING

Multi-Output Standard-Cell Mapping with MILP and Differentiable Optimization

Purpose. Improve area–delay product (ADP) by exploiting multi-output standard cells in the academic ASAP7 7-nm technology library, compared with conventional single-output mapping.

Approach. Extended the ABC logic-synthesis tool in C++ with a custom multi-output (MO) cut-discovery path and Boolean matching. Applied area-aware mixed-integer linear programming (MILP) to select matched cut/cell pairs, then passed the selected candidates to a PyTorch-based differentiable mapping optimizer for Liberty-based area/delay evaluation.

Evidence / Status. Validated correctness and evaluated 30 arithmetic benchmarks. The updated MILP-enhanced mapping flow achieved a 6% average ADP improvement over the single-output baseline.

Relevant skills: C++, ABC, logic synthesis, technology mapping, Boolean matching, PyTorch, Liberty, ASAP7, QoR analysis

ABC / MO Cut Discovery ⟶ Boolean Matching ⟶ Area-Aware MILP Selection ⟶ Differentiable Mapping